

Open-Source Electronic Signature Integration Playbook

Blue Ridge Stream Restoration & Mitigation LLC

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Ultimate KPI: Fund world-class fly fishing trip to Argentina!

1. Executive Summary

To execute stream mitigation landowner agreements (such as the **File 16 Anderson Creek Joint Venture**) and credit sales contracts with zero recurring licensing overhead, Blue Ridge Stream Restoration & Mitigation LLC can deploy **Docuseal** or **OpenSign**—the leading enterprise-grade, fully open-source alternatives to DocuSign. This playbook provides Hunter Morris with a complete engineering blueprint to deploy, configure, and operate a self-hosted e-signature platform. Additionally, it details the legal framework governing electronic signatures in the State of Georgia to ensure our contracts are 100% legally binding under both federal and state laws.

2. Open-Source Alternatives to DocuSign

DOCUSEAL DEPLOYMENT PIPELINE

[Developer Local]	■■■■■■■■■■>	[self-Hosted Server]	■■■■■■■■■■>	[Secure e-Sign]
cooperative JV		Docker / Vercel / Render		100% Legally Binding
Contract PDF		Docuseal Platform		Zero Monthly Fees

A. Docuseal (Recommended)

Docuseal is a highly polished, fully featured open-source platform designed to create, send, and execute documents electronically. It provides a drag-and-drop fields manager identical to DocuSign.

- **GitHub Repository:** <https://github.com/docusealco/docuseal>
- **Official Site:** <https://www.docuseal.com>
- **Key Advantages:**
 - **Self-Hosted for Free:** Can be deployed on a free-tier hosting provider (e.g., Render, Railway, or Fly.io) or a local machine.
 - **Custom Form Creator:** Drag-and-drop signature, initials, date, text, and checkbox fields directly onto our contract PDFs.
 - **API-First Architecture:** Allows automated landing page form submissions (like our Landowner Sourcing Form) to generate contracts instantly.

- **Zero Limits:** Send unlimited documents to unlimited signers with zero monthly fees.

B. OpenSign

OpenSign is another powerful, secure open-source e-signature platform focused on compliance, audit trails, and document security.

- **GitHub Repository:**
<https://github.com/OpenSignLabs/OpenSign>
- **Official Site:** <https://www.opensignlabs.com>
- **Key Advantages:**
- **Cryptographic Verification:** Generates PDF certificates with unique cryptographic hashes for every completed signature.
- **Comprehensive Audit Logs:** Tracks signers' IP addresses, emails, timestamps, and browser configurations.
- **Multi-Tenant Architecture:** Supports different organizational teams (e.g., land sourcing vs. credit sales).

3. Step-by-Step Self-Hosting Deployment Guide

Hunter can deploy **Docuseal** in under 10 minutes using **Docker** or free-tier cloud platforms.

Method A: One-Click Cloud Deployment (Railway or Render)

1. Create a free account on **Railway** (<https://railway.app>) or **Render** (<https://render.com>).
2. Click **New Project** and search for the pre-configured **Docuseal** template in their template registry.
3. Fill in the environment variables (keep defaults) and click **Deploy**.
4. Railway/Render will compile the image and provide a secure, private URL (e.g., <https://blueridgesign.up.railway.app>).
5. Open the URL, create your master admin account, and upload your logo to brand the interface!

Method B: Local or Virtual Private Server (VPS) Docker Deployment

If you own a dedicated server or VPS (such as a DigitalOcean droplet), deploy Docuseal using Docker Compose:

```
version: '3.8'

services:
  docuseal:
    image: docusealco/docuseal:latest
    ports:
      - "3000:3000"
    environment:
      - DATABASE_URL=postgres://user:password@db:5432/docuseal
      - PORT=3000
    volumes:
      - docuseal_data:/var/lib/docuseal
    restart: always
```

```
volumes:
  docuseal_data:
```

Run the container:

```
docker-compose up -d
```

Access the platform at [http://\[your-server-ip\]:3000](http://[your-server-ip]:3000).

4. Contract Template & Signer Mapping

Once Docuseal is running, Hunter should configure the **Roya's Cabin Anderson Creek Mitigation Agreement** template:



1. **Generate Contract PDF:** Save the finalized **File 16** (**16_Roya_Cabin_Anderson_Creek_Mitigation_Agreement.pdf**) to your computer.
2. **Upload to Docuseal:** Click **Templates > Create Template** and upload the PDF.
3. **Define Signer Roles:**
 - *Role 1:* **Landowner** (Roya Irvani)
 - *Role 2:* **Developer** (Hunter Morris / Hadi Irvani)
4. **Drag-and-Drop Fields:**
 - Scroll to the signature block page.
 - Drag a **Signature Field** and assign it to the *Landowner*. Add a **Date Field**.
 - Drag a **Signature Field** and assign it to the *Developer*. Add a **Date Field**.
5. **Publish Template:** Save the template as "Anderson Creek Mitigation JV Agreement".
6. **Send for Signature:** Click **Send**, enter Roya's email (**roya.irvani@example.com**) and Hunter's email (**bookings@flyfishingnorthgeorgia.com**), and click **Send**. The platform will securely email both parties unique links to sign.

5. Electronic Signature Legal Compliance (State of Georgia)

To ensure our joint-venture agreements and credit sales contracts hold 100% legal weight in a court of law, DocuSign/OpenSign must satisfy the strict standards of both federal and state electronic signature statutes:

A. The Federal ESIGN Act (15 U.S.C. Chapter 96)

Enacted in 2000, this federal statute grants electronic signatures the exact same legal status as traditional wet-ink pen signatures, stating that a contract "may not be denied legal effect, validity, or enforceability solely because an electronic signature was used in its formation."

B. The Georgia Uniform Electronic Transactions Act (UETA) (O.C.G.A. § 10-12-1 et seq.)

Georgia's state-level statute directly reinforces the federal ESIGN Act, confirming the validity of electronic records and signatures. To fully comply with Georgia UETA, our e-signature pipeline must satisfy four core legal pillars:

1. **Intent to Sign:** The signer must demonstrate clear intent to execute the contract. Selecting a pre-drawn signature or drawing on a canvas and clicking "Agree & Sign" legally establishes this intent.
2. **Consent to Do Business Electronically:** The platform must present a standard disclosure before signing (e.g., "By clicking agree, you consent to execute this stream mitigation agreement electronically").
3. **Association of Signature to Record:** The cryptographic signature data must be securely locked and hardcoded directly into the resulting PDF file, preventing modifications to the contract text post-execution.
4. **Retention & Access:** The system must automatically email a completed, flattened PDF copy containing the signature stamps to both the Landowner and Developer for their corporate files.

6. Sourcing Siting Economics & e-Signature Checklist

When Hunter meets a potential landowner to secure a new stream restoration reach, he can execute this 5-step sourcing process to ensure mutual confidence:

1. **Run Siting Calculator:** Proactively model the stream reach length using the operations portal calculator (e.g., a \$220,000 budget for a 1,500-LF reach yields **12,900 credits** valued at **\$2.32M**).
2. **Present 70/30 JV Economics:** Walk the landowner through the project's financials. Emphasize that they receive **30% of the credits (\$696,600 value)** with **zero financial risk**, while we absorb all CapEx and engineering liabilities.
3. **Confirm Easement Boundaries:** Align on the perpetual 100-foot buffer boundaries and explain the long-term conservation benefits for trout populations.
4. **Select Signature Method:**
 - *Method A (On-Site):* Open the **Digital Contract Signer** tab directly in our Operations Portal on a tablet, review the terms with the landowner, and draw signatures instantly.
 - *Method B (Remote):* Email a secure e-signature packet via your self-hosted **Docuseal** platform.
5. **File with USACE:** Immediately forward the signed PDF contract to Mitigation Branch Lead **Justin Hammonds** (justin.a.hammonds@usace.army.mil) as the legal binding land asset baseline for our Draft Prospectus filing.